Sean Luka Girgis

Lead Data Engineer | AWS Lakehouse Migration, Spark/PySpark, SQL & Data Quality

214-315-2190 | seanlgirgis@gmail.com

LinkedIn | GitHub | Portfolio

Professional Summary

Lead Data Engineer with 20+ years of enterprise technology experience and strong hands-on background in AWS data platforms, Python, SQL, Spark/PySpark, ETL/ELT, on-prem to cloud migration, data reconciliation, data quality validation, Hadoop/Hive concepts, Parquet-based lakehouse patterns, and production reporting systems. Experienced refactoring legacy reporting and data workflows into scalable cloud analytics patterns using S3, Glue, Athena, Redshift, Oracle, SQL optimization, partition-aware design, and documented validation processes. Best aligned to AWS lakehouse migration roles requiring practical delivery, stakeholder validation, pipeline troubleshooting, and trustworthy data products; Snowflake, Apache Iceberg, Kafka, Kubernetes, Avro, and Sybase IQ are positioned as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, and forecasting.
- Designed AWS-based data platform components using S3 landing zones, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable transformation, querying, and stakeholder-facing data products.
- Migrated heavy forecasting and analytics workloads from on-prem Oracle patterns toward AWS/Redshift-backed processing, improving scalability and enabling cloud-based reporting workflows.
- Refactored legacy manual reporting workflows into repeatable ETL processes with documented extraction logic, validation checkpoints, and operational support steps.
- Built data reconciliation and quality validation practices across BMC TrueSight, CA Wily, AppDynamics, Oracle, and AWS-backed reporting layers to improve trust in pipeline outputs.
- Applied Parquet, partitioning concepts, historical retention, schema design, and SQL optimization patterns to improve large-scale data processing and reporting access.
- Translated business and infrastructure requirements into technical designs, SQL reporting structures, dashboards, and curated data layers for technical and non-technical stakeholders.
- Troubleshot production data pipeline issues, validated outputs, documented runbooks, and coordinated with cross-functional teams for sign-off and operational readiness.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Developed analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and communicate findings to engineering teams.
- Built dashboards and operational reporting views that converted raw monitoring data into actionable performance and reliability insights.

- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and data/reporting needs.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Managed enterprise telemetry environments with 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting reporting, monitoring, and operational analytics.
- Built Perl and Korn shell automation for extracting, transforming, and distributing performance data, replacing manual exports with repeatable reporting workflows.
- Designed dashboards, alert logic, SLA thresholds, technical documentation, and reusable reporting standards adopted across client environments.
- Partnered with J2EE, WebLogic, infrastructure, database, and operations teams to interpret data, troubleshoot bottlenecks, and improve production visibility.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed enterprise telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection constraints.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, design findings, and release-readiness thresholds used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Designed and optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, Korn Shell, C++, Java

Flagship Projects

Serverless Lakehouse Platform (AWS)
Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for

raw ingestion, transformation, curated analytics datasets, query optimization, and reporting access.
Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

HorizonScale — AI Capacity Forecasting Engine
Built a forecasting-oriented analytics workflow using Python, PySpark, Prophet, scikit-learn, and Streamlit to process infrastructure telemetry, prepare model-ready datasets, and deliver capacity insights.
Technologies: Python, PySpark, SQL, Prophet, scikit-learn, Streamlit

AI-Powered Job Search Pipeline
Built a Python automation pipeline with semantic duplicate detection, LLM-based job scoring, JSON artifact generation, quality gates, document rendering, and application tracking using FAISS, sentence-transformers, and LLM APIs.
Technologies: Python, FAISS, Sentence Transformers, LLM APIs, JSON, Automation